

Metric Category	Detail	Value/Notes
Total Runtime	Application runtime	~980 ms per process (range: 974–984 ms)
MPI_Init Calls	Number	1 call per rank
MPI_Finalize Calls	Number	1 call per rank
MPI_Allreduce	Number (total/avg per node)	92 total; 92 per node
	Avg Message Size	~8.5 doubles (~68 bytes)
	Latency	~69 ns per call
MPI_Bcast	Number	2 calls per rank
	Message Size	~28 bytes
	Latency	~1323 ns per call
MPI_Reduce	Number	2 calls per rank
	Message Size	~8 bytes (2 doubles)
	Latency	~450 ns per call
MPI_Allgather	Number	3 calls per rank
	Message Size	~12 bytes
	Latency	~243 ns per call
MPI_Barrier	Number	99 calls per rank
	Latency	~75 ns per call
MPI_Send	Number	105 calls per rank
MPI_Irecv	Number	105 calls per rank
MPI_Wait	Number	105 calls per rank
Mesh Size	Dimensions	2x2 (4 ranks/nodes)
Neighbor Ranks Example	For rank 0	Ranks 1, 2, 3
Message Size Breakdown	Per neighbor	~409 bytes, ~82 bytes, ~490 bytes